3-Month Data Engineering Roadmap (Beginner → Job Ready)

🧭 Overall Plan

- **Month 1 → Foundation (Python + SQL + Basics)**
- **Month 2 → Core Data Engineering (ETL + Warehousing + Cloud)**
- **Month 3 → Advanced + Projects + Interview Prep**

📅 MONTH 1 — Foundation (Weeks 1–4)

🎯 Goal: Build strong basics (must for interviews)

✅ Week 1: Python Basics (Data Focus)

Learn only what is needed for Data Engineering.

Topics:

- Variables, Data Types
- Lists, Tuples, Dictionaries
- Conditions (if-else)
- Loops (for, while)
- Functions

Practice:

- Write programs to:
    - Count words in a file
    - Filter data from list
    - Simple data transformations

Tools:

- Python (Google Colab / VS Code)

✅ Week 2: Advanced Python for Data

Topics:

- File handling (CSV, JSON)
- Exception handling
- Modules & Packages
- Intro to Pandas

Practice:

- Read CSV → clean data → save output
- Transform JSON data

✅ Week 3: SQL (Very Important 🔥)

Topics:

- SELECT, WHERE, GROUP BY
- JOINs (Inner, Left, Right)
- Aggregations
- Subqueries

Practice:

- Use sample data (Snowflake / MySQL)
- Solve:
  - Top customers
  - Sales aggregation
  - Join multiple tables

✅ Week 4: Advanced SQL

Topics:

- Window Functions (ROW_NUMBER, RANK)
- CTEs
- Case statements

- Performance basics

Practice:

- Write real scenarios:
    - Latest record per customer
    - Running totals

📅 MONTH 2 — Core Data Engineering (Weeks 5–8)

🎯 Goal: Understand how real pipelines work

✅ Week 5: Data Engineering Basics

Concepts:

- What is ETL / ELT
- Batch vs Streaming
- Data pipeline architecture

Learn Tools:

- Apache Airflow (concept)
- Snowflake (basics)

✅ Week 6: Data Warehousing

Concepts:

- Fact vs Dimension tables
- Star Schema
- Slowly Changing Dimensions (SCD)

Practice:

- Design a small data model:
    - Sales Data Warehouse

✅ Week 7: ETL Pipeline (Hands-on)

Build mini project:

- Extract → CSV
- Transform → Python/Pandas
- Load → Snowflake

Tools:

- Python
- Snowflake

✅ Week 8: Orchestration + Automation

Topics:

- DAGs concept
- Scheduling jobs
- Dependencies

Practice:

- Build simple DAG in:
    - Apache Airflow

📅 MONTH 3 — Advanced + Real-World (Weeks 9–12)

🎯 Goal: Become job-ready 🚀

✅ Week 9: Big Data Basics

Topics:

- What is distributed computing
- Spark basics

Tools:

- Apache Spark
- PySpark basics

✅ Week 10: Cloud Data Engineering

Pick one cloud (Azure recommended for you)

Learn:

- Azure Data Factory
- Storage (Blob / ADLS)
- Basic pipeline creation

✅ Week 11: Build End-to-End Project 🔥

Project Idea:
👉 Sales Data Pipeline

Flow:

- Source → CSV/API
- Ingestion → ADF
- Transformation → PySpark / SQL
- Storage → Snowflake
- Monitoring → Airflow

✅ Week 12: Interview Prep + Optimization

Focus:

- SQL (advanced scenarios)
- Python coding
- Data modeling questions
- Project explanation

🛠️ Tools Summary

Must Know:

- Python
- SQL
- Snowflake

Good to Know:

- Apache Airflow
- Apache Spark
- Azure Data Factory

📌 Daily Study Plan (Simple)

- ⏱️ 2–3 hours/day
- 60% Practice
- 40% Learning

DM for Customised Roadmap